METRIC

MIL-STD-1877B(AT)
29 November 1991
SUPERSEDING
(see 6.4)

# MILITARY STANDARD

# BRAZING, NICKEL, HIGH TEMPERATURE VACUUM (METRIC)




AMSC N/A AREA THJM

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## FOREWORD

1. This Military Standard is approved for use by U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

3. The intent of this standard is to establish the brazing requirements in a vacuum atmosphere furnace using nickel base brazing alloys.



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# 1. SCOPE

1.1 Scope. This standard covers the requirements for brazing in a vacuum atmosphere furnace using nickel base brazing alloys.

1.2 Classification. Braze joints shall be classified as follows:

Class 1 - Joints subjected to high stresses or fatigue loading.
Class 2 - Joints subjected to intermediate stresses.
Class 3 - Joints subjected to low stresses.



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## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-H-6875 | - Heat Treatment of Steel, Process for. |
| MIL-T-81533 | - Trichlorethane, 1.1.1 (Methyl Chloroform) Inhibited, Vapor Degreasing. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-865 | - Selective (Brush Plating), Electrodeposition. |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Navy Publications and Printing Service Office, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PURCHASE DESCRIPTIONS

MILITARY

| | |
|---|---|
| M1PD-M-6061 | - Metal Texturing, Process for (Liquid Honing). |
| M1PD-N-6439 | - Nickel Plating, Process for. |
| M1PD-B-3872 | - Brazing Alloy, Nickel. |
| M1PD-B-3870 | - Brazing Alloy, Nickel. |

(Copies of M1PD Specifications and Standards may be obtained from the U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48937-5000.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).



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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

| | |
|---|---|
| AMS 4776 | - Brazing Filler Metal, Nickel Alloy. |
| AMS 4777 | - Brazing Filler Metal, Nickel Alloy |
| AMS 4779 | - Brazing Filler Metal, Nickel Alloy. |
| AMS 5510 | - Steel Sheet, Strip, and Plate, Corrosion and Heat Resistant. |

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN WELDING SOCIETY

| | |
|---|---|
| AWS A2.4 | - Standard Symbols for Welding, Brazing and Nondestructive Examination. |
| AWS A3.0 | - Welding Terms and Definitions Including Terms for Brazing, Soldering Thermal Spraying and Thermal Cutting. |

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, Florida 33135.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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## 3. DEFINITIONS

3.1 Terms. Definitions of terms relative to the equipment, processes and procedures described in this standard can be found in the AWS A2.4 and AWS A3.0 reference documents.



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## 4. GENERAL REQUIREMENTS

4.1 Equipment.

4.1.1 Brazing furnace. The brazing furnace shall be suitable for the intended purpose and shall be capable of maintaining the following conditions:

a. A vacuum of less than 0.67 Pascal (Pa) (5 microns) of mercury (Hg) at 1205 degrees Celsius (°C) [2200 degrees Fahrenheit (°F)].
b. Introducing and circulating cooling gas (argon or nitrogen) at 927°C (1700°F).
c. Uniform heating and cooling of the work load.
d. Uniform temperature in all areas of the work zone in accordance with MIL-H-6875.
e. The leak-up rate of the furnace at ambient temperature and starting at a pressure of 0.13 Pa (1 micron) of Hg or less, shall not exceed 6.67 Pa (50 microns) of Hg for a period of not less than one hour.

4.1.1.1 Alternate hydrogen atmosphere furnace. As an alternate to the vacuum furnace, a hydrogen atmosphere furnace may be used when approved in writing by the acquisition activity.

4.1.2 Instrumentation and controls.

4.1.2.1 Temperature. All brazing furnaces shall be equipped with temperature controllers and recorders in accordance with MIL-H-6875.

4.1.2.2 Vacuum gages and controls. Vacuum furnaces shall be equipped with measuring gages, emission or alphatron type or equivalent, and recording instrumentation. Their suggested location is to place them at the forward side of the main diffusion pump gate.

4.1.3 Fixtures. Brazing fixtures shall be made of non-gassing heat resistant materials, be capable of maintaining proper braze clearance, have dimensional stability, and have compatible coefficients of expansion with parts being joined.

4.1.4 Braze alloy paste applicators. Braze alloy applicator shall be capable of dispensing the required size and amount of braze alloy paste. The materials of the applicators shall not be detrimental to the alloy paste used and may be of tube, bulb, cartridge, or air gun type.

4.1.5 Air pressure source. An air pressure source providing 276 kilopascal (kPa) [40 pound force per square inch (psi)] minimum shall be required for both wet and dry abrasive blast operations.

4.2 Material.

4.2.1 Braze alloy. Braze alloy may be in any of the following forms: foil, wire, transfer tape, powder, or paste. The braze alloy paste shall be either organic vehicle or water vehicle type.



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4.2.1.1 Organic vehicle type. Organic vehicle type alloy paste shall consist of equal parts by volume of braze alloy powder (see table I) and binder. The binder viscosity may be reduced by mixing with thinner up to equal parts by volume. Additional binder and thinner may be added as required to maintain consistency during use.

Note: Preparation for use shall be accomplished by thoroughly mixing the ingredients together in a beaker cleaned with acetone.

4.2.1.2 Water vehicle type. Water vehicle type braze alloy paste shall consist of braze alloy powder and a water vehicle type binder. Premixed cartridge application paste is available from the binder manufacturer.

4.2.2 Thinner. Acetone, technical grade (99.0 percent (%) pure).

4.2.3 Braze stopoff. Braze stopoff may be either stick or liquid form and shall be capable of satisfactorily inhibiting the braze alloy wetting the surface to which the braze stopoff has been applied.

4.2.4 Cooling gas.

4.2.4.1 Argon. The purity shall be 99.995% pure with a dew point lower than -51°C (-60°F).

4.2.4.2 Nitrogen gas. Nitrogen may be substituted for argon. The purity shall be 99.995% nitrogen with a dew point of -58°C (-72°F).

4.2.4.3 Hydrogen gas. 99.95% pure with a dew point lower than -51°C (-60°F).

4.2.5 Vapor degreaser solvent. Stabilized 1-1-1 trichloroethane MIL-T-81533 or equivalent type solvent.

4.2.6 Abrasive wheels. Silicon carbide impregnated cloth wheels.

4.2.7 Wire brush. The wire brush shall have austenitic stainless steel bristles.

4.2.8 Cleaning solvent. Acetone, methyl ethyl ketone (MEK) or isopropyl or methyl alcohol of commercial grade.

4.2.9 Abrasive for abrasive blast.

4.2.9.1 Dry. Nicrobraz blasting grit - 60 or equivalent.

4.2.9.2 Wet abrasive slurry. The wet abrasive slurry shall be in accordance with MIPD-M-6061.

Note: Aluminum oxide grit shall not be considered an equivalent material.

4.2.10 Heat shields. Heat shields shall be made from AMS 5510 metal or equivalent material or fiber-frax cloth.



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4.3 Responsibility for inspection. The contractor is responsible to conduct complete and sufficient inspection to ensure:

a. Compliance with all of the requirements in this standard
b. That adequate traceability exists to verify that the requirements of this standard have been met.
c. That the manufacturing facility/supplier has the capability to braze the item in conformance with the process procedures/ techniques of this standard. This inspection shall be conducted annually. If there is more than one facility, each facility shall be subject to this inspection.



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## 5. DETAILED REQUIREMENTS

5.1 Production in process control. The contractor shall ensure that the manufacturing facility/supplier identifies the part number of the brazed item to the following (see 6.3):

a. Engineering drawing and revision.
b. Quantity of items in the lot.
c. Lot (brazing heat) number.
d. Applicable issue and revision of this standard.

5.2 Required procedures and operations.

5.2.1 Preparation of parts. All surfaces to be brazed shall be prepared in the following manner except for surfaces that are to be nickel plated (see 5.2.1.6).

5.2.1.1 Cleaning. Part surfaces shall be thoroughly cleaned using appropriate chemical solutions (see 4.2.8).

5.2.1.2 Polishing. Mating surfaces shall be prepared for brazing by polishing with a silicon carbide impregnated cloth wheel or wire brush (see 4.2.6 and 4.2.7).

Note: Polishing is not applicable to plated or otherwise specially prepared surfaces.

5.2.1.3 Vapor degrease. When required to maintain cleanliness, parts and fixtures to be placed in brazing furnace shall be vapor degreased (see 4.2.5).

5.2.1.4 Abrasive blast. All surfaces to be brazed shall be either dry or wet abrasive blasted as follows:

a. Dry abrasive blast. Dry abrasive blasting shall be accomplished using the specified material and air pressure (see 4.1.5 and 4.2.9.1).

Note: After dry abrasive blasting, parts shall be handled in a manner which will prevent contamination of the surfaces.

b. Wet abrasive blast. Except for slushing in oil, wet abrasive blasting shall be performed in accordance with all the requirements of MIPD-M-6061 (see 4.2.9.2).

5.2.1.5 Final wipe. Wet abrasive blasted parts shall have the areas to be brazed, wiped with a clean, lint free wiper wet with a suitable solvent (see 4.2.8).



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5.2.1.6 Nickel plate. Nickel plate shall be required on the braze joint area of stainless or corrosion resistant steels and superalloy parts having the equal or larger percentage by weight of the following alloying elements:

a. Titanium - 0.70%.
b. Aluminum - 0.40%.
c. Titanium plus aluminum - 0.70%.

5.2.1.6.1 Application. Nickel plate 0.010 to 0.020 millimeter (mm) (0.4 to 0.8 mil) thick shall be applied in accordance with the requirements of MIL-STD-865 or M1PD-N-6439 and the following:

a. The nickel plate shall extend 2.54 mm (0.1 inch) minimum beyond the braze joint areas indicated by the braze symbol on the engineering drawing unless part geometry does not permit this allowance.

b. The nickel plate shall be applied to each joint surface or surfaces immediately before the braze cycle and to the required braze joint surface or surfaces for each subsequent braze cycle.

Note: The nickel plating does not apply to rebraze cycles.

5.2.1.6.2 Cleaning. The nickel plated braze joint surface shall be thoroughly wiped with a suitable solvent (see 4.2.8).

5.2.2 Assembly. The detail parts shall be fixtured in the position specified on the assembly drawing.

5.2.2.1 Braze joint clearance. The optimum joint clearance between mating surfaces shall be 0.050 to 0.127 mm (0.002 to 0.005 inch ) with the following exceptions:

a. When one of the mating surfaces is knurled [0.076 to 0.127 mm (0.003 to 0.005 inch) deep], a minimum gap clearance is not required.
b. Isolated areas of the braze gap may have clearances up to 0.254 mm (0.010 inch).
c. Braze gaps at areas with sharp radii (eg, leading and trailing edges of vane to shroud joints) may have clearances up to 0.635 mm (0.025 inch).

Note: When consistent with good manufacturing practices, prick punches shall be permitted in the area to be brazed to maintain the required gap between mating surfaces.

5.2.3 Heat treatment. When heat treatment is specified on the engineering drawing, heat treatment may be combined with the braze cycle.

5.2.4 Brazing. The required class of braze joint (see 1.2.1) shall be as specified in the tail of the brazing symbol on the engineering drawing. Braze symbols shall be interpreted in accordance with AWS A2.4.



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5.2.4.1 Braze alloy paste application. Braze alloy paste (see 4.2.1.1 and 4.2.1.2) application shall be as follows:

a. Braze alloy paste shall be applied to the braze joint areas designated on the engineering drawing. The fillet shall be kept to a minimum to prevent excessive braze flow and erosion of base metal.
b. When necessary, braze stopoff (see 4.2.3) shall be applied to restrict flow of braze alloy.
c. Excess braze alloy paste shall be removed from the joints.
d. The braze alloy paste shall be dried at room temperature for 2 hours minimum or 71 to 121°C (160 to 250°F) for 50 to 70 minutes.

5.2.4.2 Braze cycle. The braze cycle shall be as follows:

a. The assembly shall be placed in the furnace (see 4.1.1). Suitable high temperature shields (see 4.2.10) shall be used as required on coated parts to protect coating from damage by direct radiation.
b. The cooling gas line shall be purged and the furnace vacuum pumped down to 1.33 Pa (10 microns) of Hg or less prior to application of heat.
c. The heating rate shall not exceed 56°C (100°F) per minute from room temperature to 427°C (800°F) and 28°C (50°F) per minute from 427°C (800°F) to the approved equalizing temperature (see table I).
d. If the pressure exceeds 6.67 Pa (50 microns) of Hg during the heat-up below 538°C (1000°F), the heat shall be turned off to allow the furnace vacuum to recover to 1.33 Pa (10 microns) of Hg or less.
e. The equalizing temperature shall be maintained until the pressure recovers to 0.27 Pa (2 microns) of Hg or less and then continued for a period of 60 minutes minimum.
f. The heat-up rate from the equalizing temperature to the brazing temperature shall be at a specified control rate not to exceed 2.2°C (5°F) per minute.
g. Unless otherwise specified, the assembly shall be maintained at the specified brazing temperature (see table I) for 10 to 30 minutes.
h. The heat-up rate, equalizing holding time, and brazing temperature and time shall be as approved by the acquisition activity.
i. Unless otherwise specified on the engineering drawing, assembly shall be cooled in a vacuum, argon gas, or nitrogen gas atmosphere. If an argon or nitrogen gas is used, not more than 50.8 mm (2 inches) gage pressure of cooling gas shall be introduced until reaching a part temperature of 204°C (400°F).
j. Furnace door may be opened below 204°C (400°F).



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5.2.4.2.1 Hydrogen atmosphere braze cycle. Request by a contractor to use the alternate hydrogen atmosphere brazing cycle must be submitted in writing for approval. Data to be included shall contain full processing procedure to the acquisition activity.

5.2.4.3 Multiple cycles. Application of braze alloy paste and the braze cycle (see 5.2.4.1 and 5.2.4.2) shall be repeated as required for assemblies requiring more than one braze cycle to complete the assembly operation.

5.3 Monitoring equipment.

5.3.1 Accuracy of furnace control instruments. The accuracy of temperature measuring and controlling instruments shall be checked at regular intervals of at least once in every 90 working days in accordance with the requirements of MIL-H-6875. The calibrated thermocouple shall be of the platinum type or chromel/alumel type.

5.4 Test methods.

5.4.1 Inspection. The method of braze inspection and acceptance levels shall be in accordance with the applicable inspection specifications cited on the engineering drawing.



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## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard is intended to define the requirements for brazing in a vacuum atmosphere furnace using nickel base brazing alloys.

6.2 Issue of DODISS. When this standard is used in acquisitions, the applicable issue of the DODISS must be cited in the solicitation (see 2.1.1 and 2.2).

6.3 Data requirements. Although this standard does not include any requirements for data, it does not exclude contractors from obtaining data from its suppliers as required to ensure compliance to this standard and to verify conformance to the applicable engineering drawings.

6.4 Supersession data. This standard supersedes MIL-STD-1877A(AT), dated 7 June 1985 and Textron Lycoming specification P6231M, dated 12 December 1990.

6.5 Subject term (key word) listing.

| | |
|---|---|
| Abrasive blast | Methyl/Ethyl Ketone (MEK) |
| Aceton | Nickel plate |
| Alphatron | Silicon carbide |
| Aluminum | Thinner |
| Braze alloy powder | Titanium |
| Diffusion pump gate | Vacuum furnace |
| Hydrogen | Vapor degrease |
| Isopropyl | Viscosity |

6.6 Changes from previous issue. Vertical lines or asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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TABLE I. Brazing alloy, and reference brazing temperature.

| Braze alloy | Equalizing temperature (1) | Brazing temperature (1) |
|---|---|---|
| AMS 4776 (2) | 949°C (1740°F) | 1121-1135°C (2050-2075°F) |
| AMS 4777 | 949°C (1740°F) | 1038-1066°C (1900-1950°F) |
| M1PD-B-3872 (2)<br>M1PD-B-3870 | 949°C (1740°F) | 1154-1170°C (2110-2135°F)<br>(3) |
| AMS 4779 | 949°C (1740°F) | 1104-1121°C (2020-2050°F) |
| AMS 4782 | 949-1066°C (1740-1950°F) | 1155-1204°C (2100-2200°F) |
| AMS 4783 | 949-1066°C (1740-1950°F) | 1176-1231°C (2150-2250°F) |

Note:

(1) The temperatures noted are actual part temperatures as established in furnace calibrations.

(2) Brazing cycle for components containing coating parts must be maintained within 1121-1134°C (2050-2075°F).

(3) For aluminide coated parts, the brazing temperature range shall be 1121-1155°C (2050-2100°F).



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**NOTE:** This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. AREA THJM

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-STD-1877B(AT); Brazing, Nickel, High Temperature Vacuum (Metric)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

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C. REASON FOR RECOMMENDED CHANGE(S)

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